

ADVANCED SECONDARY CURRICULUM

Competency Based Physics

Scenario items

COMPILED BY

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ADVANCED SECONDARY PHYSICS ITEM

A technology company is rolling out smart transport cards to facilitate the daily commute of its employees. These employees travel by bus from a trading centre to the company offices located near the main bus park, five days a week.

However, the company is currently facing a challenge in determining the exact amount of money to deposit on each smart card for a full month of travel.

On one particular day, the bus journey was recorded as follows:

It started from rest at the trading centre and accelerated uniformly at 1.25 m/s^2 until it reached a speed of 72 km/h . The bus then continued at this constant speed for a distance of 50 km before decelerating uniformly at 2 m/s^2 to come to a complete stop at the bus park.

The transport service charges Ushs 1,000 per kilometre of travel.

Task:

Represent the entire journey of the bus graphically

Advise the company on the minimum amount of money that should be deposited on each smart card to cover a full month's travel expenses.

ADVANCED SECONDARY PHYSICS ITEM

As part of your school's sports science project, your team is working with the athletics coach to help a long jumper improve his performance. During training, he takes off from ground level with an initial speed of 5m/s at an angle of 30° to the horizontal.

To qualify for the upcoming district competitions, a jumper must achieve a minimum horizontal distance of 9.0 metres from the point of take-off to the landing point.

Task

Help the school administration to know whether the jumper qualifies for the competitions.

Using principles of projectile motion, explain with evidence how the jumper can increase his horizontal distance.

Derive the equation of trajectory.

ADVANCED SECONDARY PHYSICS ITEM

A company that manufactures cooking utensils plans to develop a new type of metal saucepans that heat up quickly and uses less energy. However, the company does not yet know how to determine the specific heat capacity of the metal sample they intend to use.

Knowing the SHC of the metal will help the company estimate how much energy is required to raise the temperature of their saucepans, which directly affects cooking time and fuel efficiency.

Task

Advice the company manager on the steps he can take to determine the specific heat capacity of the sample and how they can ensure accurate results.

Using the data below, calculate the specific heat capacity of the metal block:

Mass of metal block = 1.8 kg

Voltage = 12 V

Current = 2 A

Time of heating = 5 minutes

Temperature increase = 30°C

ADVANCED SECONDARY PHYSICS ITEM

Your school has recently acquired new physics lab equipment, including a continuous flow calorimeter designed to measure the specific heat capacity (SHC) of liquids. During a science club session, your group sets up the apparatus for a public exhibition. As part of the demonstration, your team is required to show visiting parents how the setup can be used to determine the SHC of a liquid, such as water.

Task As a student of physics;

Use your set up to explain to the visitors how the specific heat capacity of the liquid can be determined.

Help them understand the advantages and disadvantages of using the above method

In such a method, 50g of water is collected in 1 minute, the voltmeter and ammeter readings are 12.0V and 2.50A respectively. While the inflow and outflow temperatures are 20°C and 28°C respectively. When the flow rate is reduced to 25gmin⁻¹, the voltmeter and ammeter read 8.8V and 1.85A respectively while the temperatures remain constant. Calculate the specific heat capacity of water

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ADVANCED SECONDARY PHYSICS ITEM

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Derive the equation of trajectory.

ADVANCED SECONDARY PHYSICS ITEM

A researcher at a local science institute is studying the behavior of simple pendulums for use in timing devices. While conducting his experiment, he suspects that the time it takes for the pendulum to complete one full swing (T) depends on;

The mass (m) of the bob

The length (L) of the string

Acceleration due to gravity (g) at his testing location.

However, he does not know the exact formula connecting these variables. During the discussion with a colleague, he was told that if the pendulum moves in a circular path, the centripetal force (F) acting on the bob is given by $F = \frac{mv^2}{r}$ where m is the mass of the bob, r is the radius of the path and v is the speed of the bob. The researcher is not sure about the correctness of the formula.

Task

As a learner of physics,

Help the researcher to derive the formula for the period of a simple pendulum

Use dimensional analysis to confirm whether the formula for centripetal force is correct

Enable the researcher to understand the limitations of using dimensional analysis to derive equations

ADVANCED SECONDARY PHYSICS ITEM

Rwenzori high school is facing frequent blackouts and rising electricity bills, which disrupt lessons in the computer lab and science practicals. The head teacher Mr. James, is considering solar energy as an alternative. In a science club meeting, Sarah, the science perfect believes solar energy could sustain the school if harnessed well. But John, the debate club leader disagrees saying ‘before recommending solar, we must understand how the sun actually works. Do we know its structure, how it produces energy, and whether it really is the primary source of other renewable energies like hydro, wind, biomass, geothermal and tidal?’

The district environment officer adds that “to advise the school properly, you must also study how solar thermal systems, photovoltaic panels, and thermal storage work. Then weigh its importance and finally prepare clear recommendations for the school”

Task: As a student of physics;

- a) Explain the composition and structure of the sun and the process of energy production in it.
- b) Show how the sun is the primary source of hydro, wind, biomass, geothermal, and tidal energy
- c) Investigate the operation of solar thermal systems, photovoltaic panels and thermal storage systems.
- d) Debate the motion “solar energy is the best option for sustaining life on Earth”.

ADVANCED SECONDARY PHYSICS ITEM

In kasese district, a community is constructing a simple water purification system using narrow glass tubes and containers. During testing, they observe unusual behaviors; water rises in thin tubes, droplets form on container walls, and soap makes water spread more easily. The community wants to understand why these behaviors occur in order to improve the design of their purification system.

Tasks

As a physics student;

Explain the occurrence of surface tension and capillarity, and discuss the factors that affect them

Show how to determine the coefficient of surface tension and how to measure the angle of contact, explaining their implications on liquid behavior.

ADVANCED SECONDARY PHYSICS ITEM

You are a road safety expert investigating a bus accident. The bus was moving at 20 m/s when the driver applied the brakes, bringing the bus to a stop in 4 seconds. A passenger inside the bus was not wearing a seatbelt and continued moving forward after the bus stopped.

Task:

1. Calculate the deceleration of the bus.
2. Determine how far the passenger would travel forward before stopping, assuming no external force acts on them.
3. Explain the importance of seatbelts in preventing injuries.
4. Discuss how Newton's First Law of Motion applies to this scenario.

ADVANCED SECONDARY PHYSICS ITEM

A construction worker accidentally drops a hammer from the top of a 100 m tall building. You need to determine how long it will take for the hammer to reach the ground and its speed just before impact.

Task:

1. Calculate the time it takes for the hammer to reach the ground assuming free fall.
2. Determine the velocity of the hammer just before it hits the ground.
3. Discuss real-world factors like air resistance and how they might affect the motion.
4. Suggest safety measures that could prevent such accidents at construction sites.

ADVANCED SECONDARY PHYSICS ITEM

You are an engineer working on the construction of a bridge. The bridge's support beams must be designed to withstand the forces acting on them without bending excessively. You need to determine whether the stress equation used in the design is dimensionally consistent.

Task:

1. Identify the fundamental and derived quantities involved in stress calculation.
2. Use dimensional analysis to verify the equation for stress:

$$\sigma = \frac{F}{A}$$
 where F is the applied force and A is the cross-sectional area.
3. Explain why ensuring dimensional consistency is important in structural engineering.
4. Discuss potential problems if an incorrect formula is used and how to resolve them.

ADVANCED SECONDARY PHYSICS ITEM

You are a sports scientist analysing the motion of sprinters to optimize their performance. The equations used to calculate acceleration and force must be verified for accuracy.

Task:

1. Identify the fundamental and derived quantities involved in an athlete's motion.
2. Use dimensional analysis to check the validity of the equation: $v = u + at$ where v is final velocity, u is initial velocity, a is acceleration, and t is time.
3. Explain why verifying dimensions is necessary in sports biomechanics.
4. Discuss how incorrect calculations could affect an athlete's performance analysis.

ADVANCED SECONDARY PHYSICS ITEM

A skydiver jumps from an airplane at 4,000 meters above the ground. You need to calculate how long the skydiver will fall before deploying their parachute, assuming free fall without air resistance.

Task:

1. Calculate the time taken to fall 4,000 meters under gravity.
2. Determine the velocity just before deploying the parachute.
3. Explain how air resistance and terminal velocity affect real-world skydiving.
4. Discuss the importance of parachute design and deployment timing in ensuring safety.

ADVANCED SECONDARY PHYSICS ITEM

You are working as a design engineer at a manufacturing company. Your team is developing a new high-precision weighing scale that will be used in laboratories to measure small masses accurately. The device must be calibrated correctly to ensure precision in its measurements. Before proceeding with the design, you need to verify the consistency of the formulas used in the calculations.

Task:

1. Identify the fundamental and derived physical quantities required in the calibration of the weighing scale.
2. Use dimensional analysis to check whether the following equations used in the scale design are dimensionally consistent:
 - (i) The force acting on the mass: $F = mg$
 - (ii) The oscillation period of the scale's pan: $T = 2\pi \sqrt{\left(\frac{l}{g}\right)}$
3. Explain why dimensional analysis is crucial in verifying these equations before building the prototype.
4. Discuss any potential design challenges that could arise due to errors in dimensions and how they can be corrected.

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The mass (m) of the bob

The length (L) of the string

Acceleration due to gravity (g) at his testing location.

However, he does not know the exact formular connecting these variables. During the discussion with a colleague, he was told that if the pendulum moves in a circular path, the centripetal force (F) acting on the bob is given by $F = \frac{mv^2}{r}$ where m is the mass of the bob, r is the radius of the path and v is the speed of the bob. The researcher is not sure about the correctness of the formular.

Task

As a learner of physics,

Help the researcher to derive the formular for the period of a simple pendulum

Use dimensional analysis to confirm whether the formular for centripetal force is correct

Enable the researcher to understand the limitations of using dimensional analysis to derive equations

ADVANCED SECONDARY PHYSICS ITEM

During a fire emergency, a firefighter uses a water cannon to direct a stream of water towards a burning building. The nozzle is set to spray the water at an angle of 60° above the horizontal, with a speed of 20 m/s. The water stream travels a horizontal distance of 30 meters and just clears the roof of the building before descending on the other side. After the incident, the insurance company investigating the damage wants to know the exact height of the building.

Task

- Help the insurance company to know the height of the building
- How far beyond the building does the water land?
- Describe one real life situation where the knowledge of projectile motion is applied.

ADVANCED SECONDARY PHYSICS ITEM

As part of laboratory investigation to determine the specific latent heat of vaporisation of water, you are provided with 2 copper calorimeters, a steam trap, a thermometer and a stirrer.

From the investigation, the following results were obtained:

Mass of empty calorimeter = 150g

Mass of water added = 340g

Mass of condensed steam = 35g

Initial temperature of the calorimeter and its contents = 15°C

Temperature of steam = 100°C

Final temperature of the mixture = 71°C

Hint

Specific heat capacity of water = $4200\text{Jkg}^{-1}\text{K}^{-1}$

Specific heat capacity of copper = $400\text{Jkg}^{-1}\text{K}^{-1}$

Task

As a student of physics;

Describe how the specific latent heat of vaporisation of water can be determined using the above apparatus.

Explain how you can minimise errors in (a) above

Determine the specific latent heat of vaporisation of water

ADVANCED SECONDARY PHYSICS ITEM

During a recent trip to the beach, senior five students were engaged in different gaming activities in different groups.

One group at the swimming pool site dropped a ball into the water and saw circular ripples spreading out on the water surface.

Another group participated in swinging from three different swings attached to the same horizontal rod. To their surprise, with only one person given a push, all the rest were also set into oscillation after some time until they attained a maximum displacement before finally coming to rest. When one of them touched the horizontal rod, some heat was felt at the point of attachment with the chain.

Task: As a physics scholar in Advanced Level;

- (a) Help your friends from the Arts class to understand the following terms; an oscillation, amplitude, frequency and wavelength as applied to their activities.
- (b) Show that the velocity V , frequency, f and wavelength, λ of the ripples on the water surface are related by the equation $V=f \lambda$.
- (c) Explain to your friends why others were able to oscillate even when they had not been physically given a push.
- (d) Explain the cause of the heat at the point of attachment between the chain and the rod.
- (e)(i) State and define the phenomenon where all the three swinging students attained the same amplitude.
- (ii) Explain why their amplitude of oscillation finally came to zero.

ADVANCED SECONDARY PHYSICS ITEM

The traffic police are investigating a recent road accident involving a Toyota Premio and a bus. Witnesses reported that after the collision, the Premio moved forward at a velocity of 20ms^{-1} While the bus was seen reversing out 60ms^{-1} . According to the records, the Toyota Premio had mass of 1200kg and the bus had a mass of 6000 kg .

Task

As a student of physics;

- Help the police to know the initial velocities of both the Premio and the bus.
- Show that $U_2 - U_1 = 80\text{ms}^{-1}$, where U_1 and U_2 are the initial velocities of the Premio and the bus respectively
- Describe the real-life applications of the principal of conservation of linear momentum.

ADVANCED SECONDARY PHYSICS ITEM

During a fire emergency, a firefighter uses a water cannon to direct a stream of water towards a burning building. The nozzle is set to spray the water at an angle of 60° above the horizontal, with a speed of 20 m/s. The water stream travels a horizontal distance of 30 meters and just clears the roof of the building before descending on the other side. After the incident, the insurance company investigating the damage wants to know the exact height of the building.

Task

- Help the insurance company to know the height of the building
- How far beyond the building does the water land?
- Describe one real life situation where the knowledge of projectile motion is applied.

ADVANCED SECONDARY PHYSICS ITEM

During a science exhibition at your school, the science club plans to demonstrate how to determine the specific heat capacity of a liquid using the continuous flow method. They want to explain the procedures clearly to parent and visitors however, some of the Club members do not know the method.

	Set 1	Set 2
Voltage	35.0	26.0
Current	2.0	2.0
Mass (kg) collected in seconds	0.047	0.030
Inflow temperature	25.0	25.0
Out flow temperature	29.0	29.0

Task

As a student of physics,

- Help the Club members to understand the Continuous flow method.
- Help the students identify the advantages and disadvantages of the continuous Flow method.
- Determine the specific capacity the liquid using the table below.

ADVANCED SECONDARY PHYSICS ITEM

While studying the behaviour of charged particles, a researcher sets up an experiment where electrons are accelerated from rest and pass between 2 horizontal parallel plates that produce a uniform electric field of intensity $2.4 \times 10^3 \text{ nc}^{-1}$, he observes that as an electron leaves the field region, it follows a curved path rather than continuing straight. The researcher wants to understand the nature of this motion

Hint;

Electron charge	$= 1.6 \times 10^{-19}$
Electron mass	$= 9.11 \times 10^{-31} \text{ kg}$

Task

Help the researcher to;

- Derive an expression for the vertical deflection of the electron as it leaves the plates and show that its path is parabolic.
- Determine the force on the electron
- Determine the acceleration of the electron
- Determine the velocity acquired in moving through a p.d of 90 V.

ADVANCED SECONDARY PHYSICS ITEM

A team of engineering students is testing the safety of a 500 g emergency supply drone designed to deliver medicine. The drone is placed on a smooth surface in the lab, and four guiding cables are attached to it. To test stability, the team pulls the cables in different directions with forces of **3.0 N**, **3.5 N**, **4.5 N**, and **5.0 N**, as shown in **Figure 1** below. Initially, the drone is at rest.

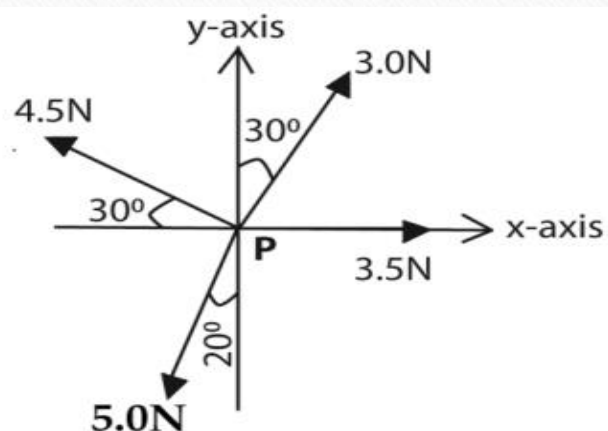


Figure 1: Forces acting on the drone

Later, while transporting the same drone inside a lift, the students notice that its weight appears to change when the lift moves. One of them even claims to feel momentarily weightless.

Task

Explain why one of the students feels weightless

Help the students to understand why the tension in the cable of a lift is not the same when it is ascending as when it is descending.

If all four forces act on the drone simultaneously, determine the distance it will move in 5 seconds.

ADVANCED SECONDARY PHYSICS ITEM

A researcher is investigating the thermal properties of a liquid. He is particularly interested in finding out how much heat is required to raise the temperature of the liquid by one degree Celsius and how much heat is required to completely vaporize it. Although he has the necessary laboratory setup, he does not know how to conduct the experiments and therefore seeks guidance.

Liquid	Inlet Temp (°C)	Outlet Temp (°C)	Flow Rate (g/min)	Current (A)	P.d (V)
Water	15.2	17.4	20.0	2.3	3.3
Oil	15.2	17.4	70.0	2.7	3.9

In his trials he passed water and oil in the continuous flow calorimeter and obtained the results above.

Hint : Specific heat capacity of water = $4200 \text{ J kg}^{-1} \text{ K}^{-1}$

Task

- Guide the researcher on how to perform the experiments
- Basing on the values in the table, determine the specific heat capacity of oil
- Explain why the specific latent heat of vaporisation is greater than that of fusion for the same substance

ADVANCED SECONDARY PHYSICS ITEM

During a physics demonstration, students are investigating the motion of charged particles in electric and magnetic fields. They set up two horizontal parallel metal plates, each 5 cm long and 4.0 cm apart, in a vacuum tube. The upper plate is connected to a potential of 400 V while the lower plate is earthed. An electron beam is projected horizontally, mid-way between the plates, with a velocity of 1.0×10^7 m/s. However, one student insists that the path of the electron should remain straight since it was projected horizontally with no vertical velocity component. Another student argues that the electron is under the influence of the electric field and should therefore be deflected. Later, the teacher uses the same tube to demonstrate cathode rays and compare them with positive rays.

In another setup, a beam of positive ions is accelerated through a potential difference of 1000 V into a uniform magnetic field of flux density 0.2 T. Inside the field, the ions move in a circular path of radius 2.3 cm.

Hint

Electron charge = 1.6×10^{-19} C

Electron mass = 9.11×10^{-31} kg

Task

Show that the path of the electron in the electric field between the plates is a parabola.

Calculate the vertical deflection of the electron as it emerges from the plates.

Help the students to know how cathode rays are produced

Enable the students to understand the differences between cathode rays and positive rays

Derive an expression for the charge to mass ratio of the ions in the magnetic field and calculate its value

ADVANCED SECONDARY PHYSICS ITEM

You are a student preparing for a science exhibition. Your project is titled “Exploring Light with Mirrors”. You decide to demonstrate how light behaves when reflected by plane and curved mirrors using simple experiments and real-life applications.

Tasks:

- (a) Begin by demonstrating the laws of reflection using a ray diagram and a plane mirror. Explain how the angle of incidence equals the angle of reflection
- (b) Rotate the plane mirror and show that if the mirror is rotated by an angle θ , the reflected ray turns by 2θ . Derive this relationship keeping the incident ray constant.
- (c) Use a model of a sextant to show how it helps in measuring the angle of elevation of a star.
- (d) Set up an experiment to determine the focal length of a convex mirror using a plane mirror. Describe the procedure clearly.
- (e) A plane mirror is placed 10 cm in front of a convex mirror so that it covers about half of the convex mirror surface. A pin placed 20 cm in front of the plane mirror gives an image which coincides with that of the pin in the convex mirror. Find the focal length of the convex mirror.

ADVANCED SECONDARY PHYSICS ITEM

The engineers at **Wave FM** are analysing the performance of their radio transmission system. The station broadcasts electromagnetic waves with a wavelength of **10 metres**. During their tests, they encounter various situations, such as free oscillations of the transmitter, damped oscillations caused by environmental interference, and forced oscillations due to external disturbances.

To ensure their broadcast quality, the engineers have approached you help them address the following questions:

- a) Calculate the **frequency** of the emitted radio wave.
- b) Determine the **period** of the wave.
- c) Compute the **number of oscillations** that occur in **10 seconds**.
- d) State any **assumptions** made during these calculations.
- e) Explain the concepts of free oscillations, damped oscillations, and forced oscillations, giving any two examples of each.
- f) Discuss any five other real-life **applications of waves**.

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Task

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Explain to your friends why others were able to oscillate even when they had not been physically given a push.

Explain the cause of the heat at the point of attachment between the chain and the rod. (2 scores)

- (i) State and define the phenomenon where all the three swinging students attained the same amplitude.
- (ii) Explain why their amplitude of oscillation finally came to zero.